

$$\lim \frac{1}{n^\alpha} = 0 \quad \alpha > 0$$

$$\lim q^n = 0 \quad |q| < 1$$

$$\lim \left(1 + \frac{r}{n}\right)^n = e^r \quad r \in \mathbb{R}$$

$$\lim a_n^{b_n} = e^{\lim (a_n - 1) b_n}$$

$$\uparrow \lim a_n = 1$$

$$\lim a_n = \pm \infty$$

$$\lim f(a_n) = f(\lim a_n) \leftarrow \text{že je } f \text{ zvezna}$$

1. a) $\lim_{n \rightarrow \infty} \frac{1+2+\dots+n}{n^2} = \lim_{n \rightarrow \infty} \frac{n(n+1) \cdot \frac{1}{n^2}}{2n^2 \cdot \frac{1}{n^2}} = \frac{1}{2}$

b) $\lim_{n \rightarrow \infty} \frac{2^n + 3^n}{2^{n+1} + 3^{n+1}} = \lim_{n \rightarrow \infty} \frac{\left(\frac{2}{3}\right)^n + 1}{2 \cdot \left(\frac{2}{3}\right)^n + 3} = \frac{1}{3}$

c) $\lim \log_2 \left(\frac{2n^2 + n + 1}{(n+1)^2} \right) = \log_2 \left(\lim \frac{2n^2 + n + 1}{n^2 + 2n + 1} \right) = \log_2 2 = 1$

d) $\lim \left(\frac{n-2}{n+1} \right)^{2n+2} = \lim \left(1 + \frac{-3}{n+1} \right)^{2n+2} = \lim \left(\left(1 + \frac{-3}{n+1} \right)^{n+1} \right)^2 = (e^{-3})^2 = e^{-6}$

e) $\lim n \left(\log_2(2n-1) - \log_2 n - 1 \right) = \lim n \left(\log_2 \frac{2n-1}{2n} \right) = \lim \log_2 \left(1 + \frac{-1}{2n} \right)^n =$
 $= \log_2 \left(\lim \left(1 + \frac{-1}{2n} \right)^{2n} \right)^{\frac{1}{2}} \stackrel{(\log_2)^2}{=} \log_2 e^{-\frac{1}{2}} = \underline{\underline{-\frac{1}{2} \log_2 e}}$

2. Trditev:

$$(a_n)_{n \in \mathbb{N}} ; a_n > 0 \text{ za vsak } n \in \mathbb{N}$$

$$\lim \frac{a_{n+1}}{a_n} = A \Rightarrow \lim \sqrt[n]{a_n} = A$$

Dokaz:

$$a_{nn} = A \cdot a_n = A^2 a_{n-1} = \dots = A^n a_1$$

$$\sqrt[n]{a_{n+1}} = \sqrt[n]{A^n a_n} = A \sqrt[n]{a_n} = A$$

$$a) \lim_{n \rightarrow \infty} \sqrt[n]{n} = \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = 1 \quad a_n = n \Rightarrow \lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$$

$$b) \lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{n!}} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{n^n}{n!}} = \lim_{n \rightarrow \infty} \frac{\frac{(n+1)^{n+1}}{(n+1)!}}{\frac{n^n}{n!}} = \lim_{n \rightarrow \infty} \frac{(n+1)(n+1)^n}{(n+1)n^n} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right) = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) = e$$

$$c) \lim_{n \rightarrow \infty} \frac{\ln n}{n} = \lim_{n \rightarrow \infty} \ln n^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \ln \sqrt[n]{n} = \ln(\lim_{n \rightarrow \infty} \sqrt[n]{n}) = \ln 1 = 0$$

Sandvič:

$$a_n, b_n, c_n$$

$$a_n \leq b_n \leq c_n$$

$$\text{Če } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = A \Rightarrow \lim_{n \rightarrow \infty} b_n = A$$

Primer

$$0 \leq \frac{\ln n}{n} = \frac{2 \ln \sqrt{n}}{n} \leq \frac{2 \sqrt{n}}{n} = \frac{2}{\sqrt{n}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$$

3) Dokaži, da je konvergentno

$$a_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}$$

$$a_{n+1} - a_n = \frac{1}{n+2} + \dots + \frac{1}{2n+1} + \frac{1}{2n+2} - \left(\frac{1}{n+1} + \dots + \frac{1}{2n} \right) = \frac{1}{2n+1} + \frac{1}{2n+2} - \frac{1}{n+1}$$

$$> \frac{2}{2n+2} - \frac{1}{n+1} = 0 \Rightarrow \text{naraščajoče}$$

- navedeno omejeno:

$$\frac{1}{n+1} + \dots + \frac{1}{2n} < \ln 2$$

$$\int_n^{2n} \frac{1}{x} dx = \ln x \Big|_n^{2n} = \ln \frac{2n}{n} = \ln 2$$

4. Rekurzivna zap.

$$a_1$$

$$a_{n+1} = f(a_n)$$

$$\downarrow \lim \quad \downarrow \lim$$

$$A = f(A) \rightarrow \text{kandidati za limito}$$

$$a_1 = 0$$

$$a_{n+1} = \sqrt[3]{7a_n + 6}$$

$$\downarrow n \rightarrow \infty$$

$$A = \sqrt[3]{7A + 6}$$

$$A^3 - 7A - 6 = 0$$

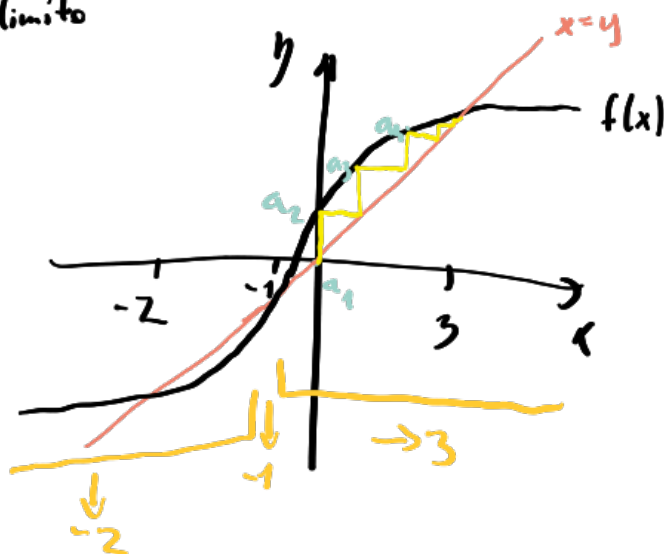
$$(A+1)(A^2 - A - 6) = 0$$

$$(A+1)(A-3)(A+2) = 0$$

	1	0	-7	-6
-1		-1	1	6
	1	-1	-6	0

$$\left. \begin{array}{l} A_1 = -1 \\ A_2 = 3 \\ A_3 = -2 \end{array} \right\} \text{kandidati za limito}$$

$$f(x) = \sqrt[3]{7x + 6}$$



$$a_1 = 0$$

$$a_2 = \sqrt[3]{6}$$

$$a_3 = \sqrt[3]{7\sqrt[3]{6} + 6}$$

nar.:

$$h=1:$$

$$\sqrt[3]{6} \geq 0 \checkmark$$

$$\text{I.P. } n \rightarrow n+1$$

$$\sqrt[3]{7a_n + 6} \geq \sqrt[3]{7a_{n-1} + 6}$$

$$7a_n + 6 \geq 7a_{n-1} + 6$$

$$a_n \geq a_{n-1} \checkmark$$

omejeno s 3:

$$a_n \leq 3 \checkmark$$

✓
1p. $a_{n+1} \leq 3$

$$\sqrt[3]{7a_n + 6} \leq 3$$

$$7a_n \leq 21$$

$$a_n \leq 3$$

✓

